

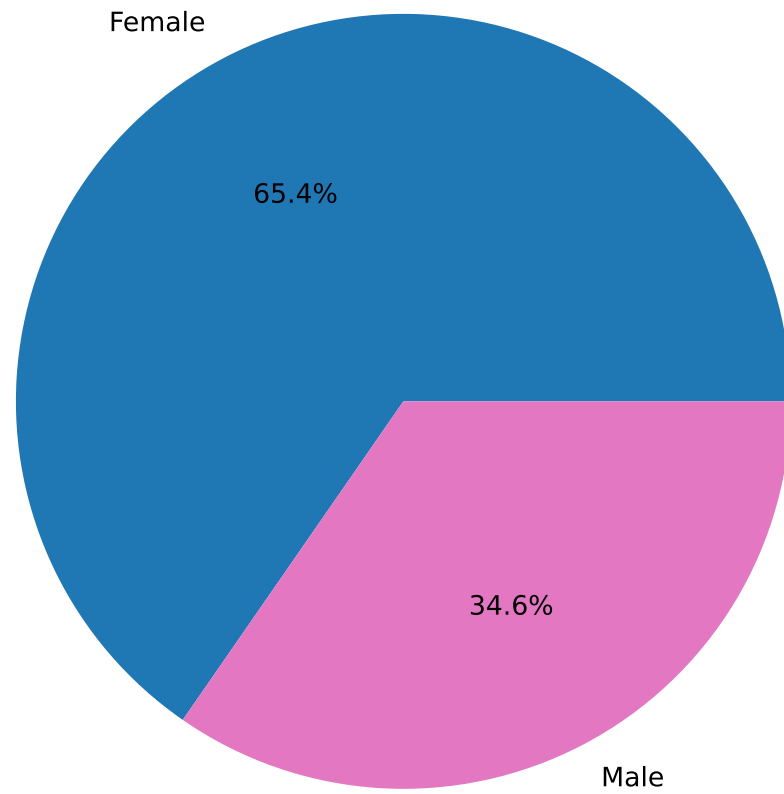
Machine Learning Using Multiparametric MRI Radiomic Feature Analysis to Predict Ki-67 in WHO Grade I Meningiomas

Implementation of Khanna et al. 2021 Methodology

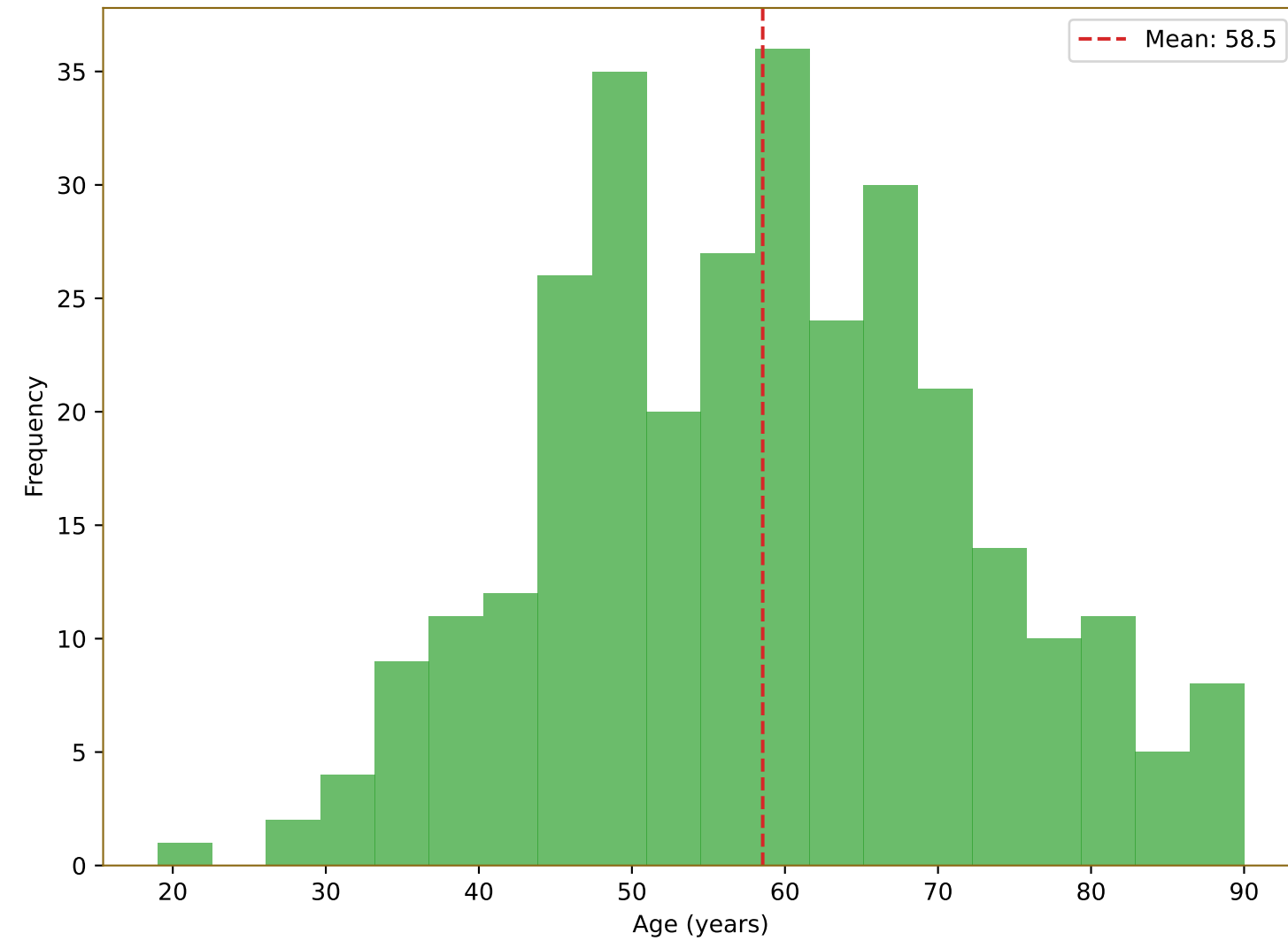
- Patient Demographics & Tumor Characteristics
- Ki-67 Distribution Analysis
- Radiomic Feature Extraction & Selection
- Machine Learning Model Training (LASSO + SVM)
- Model Performance Evaluation
- Feature Importance Analysis [306 patients, 2551 radiomic features](#)
- Clinical Implications & Recommendations

Patient Demographics & Tumor Characteristics

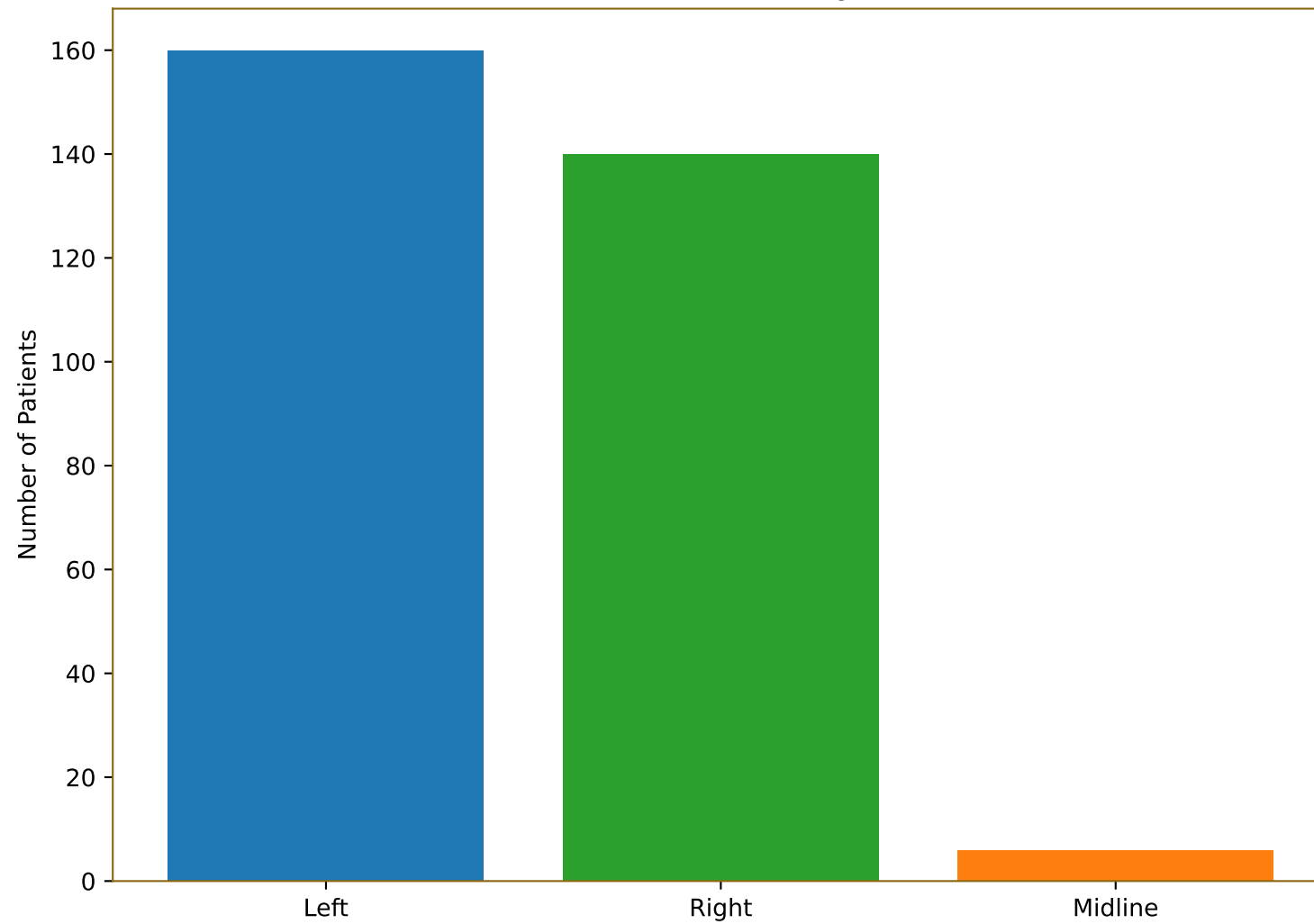
Gender Distribution



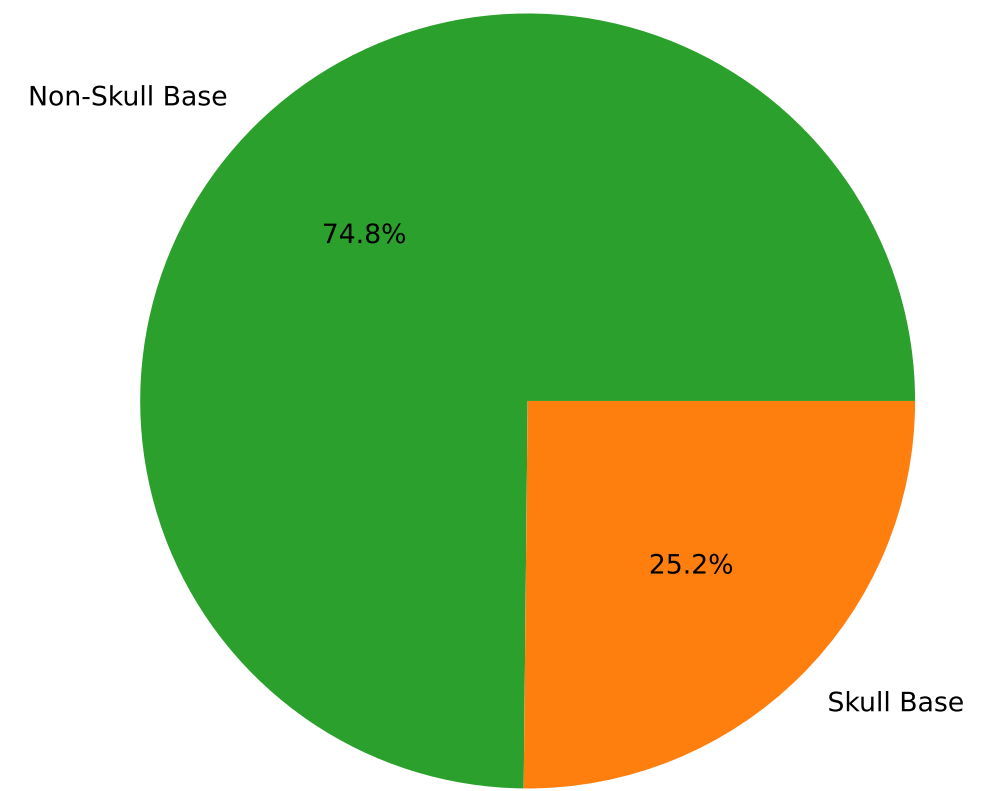
Age Distribution



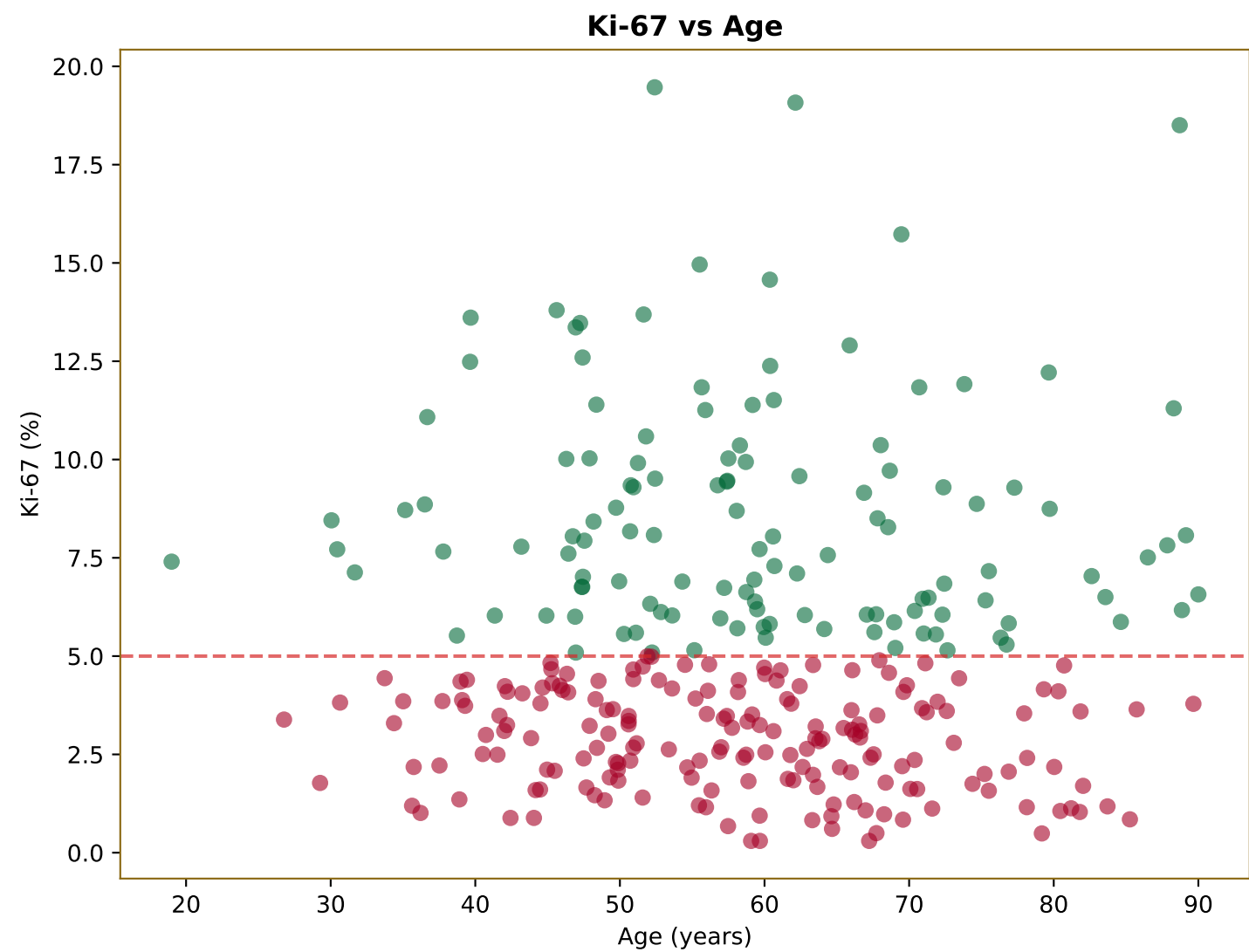
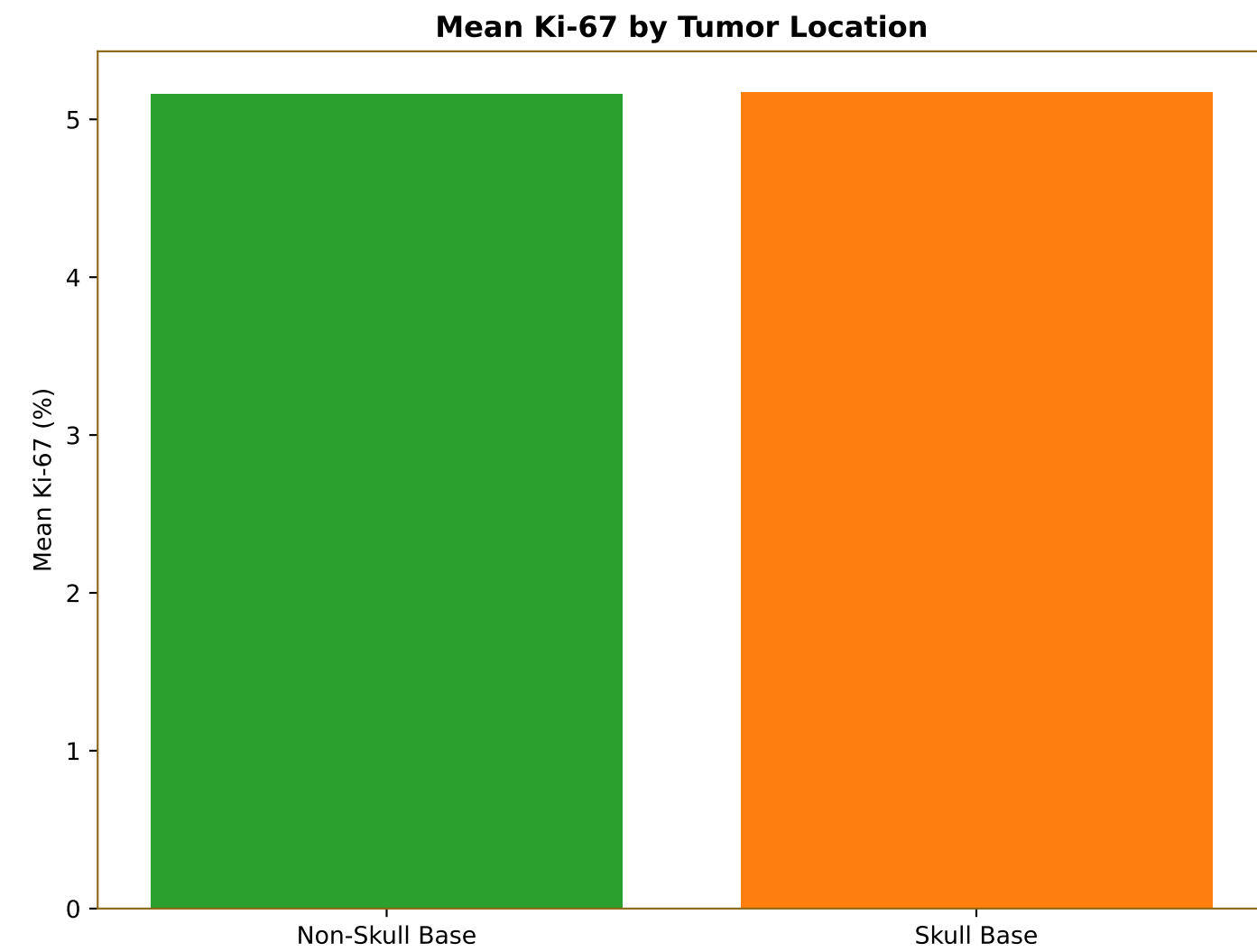
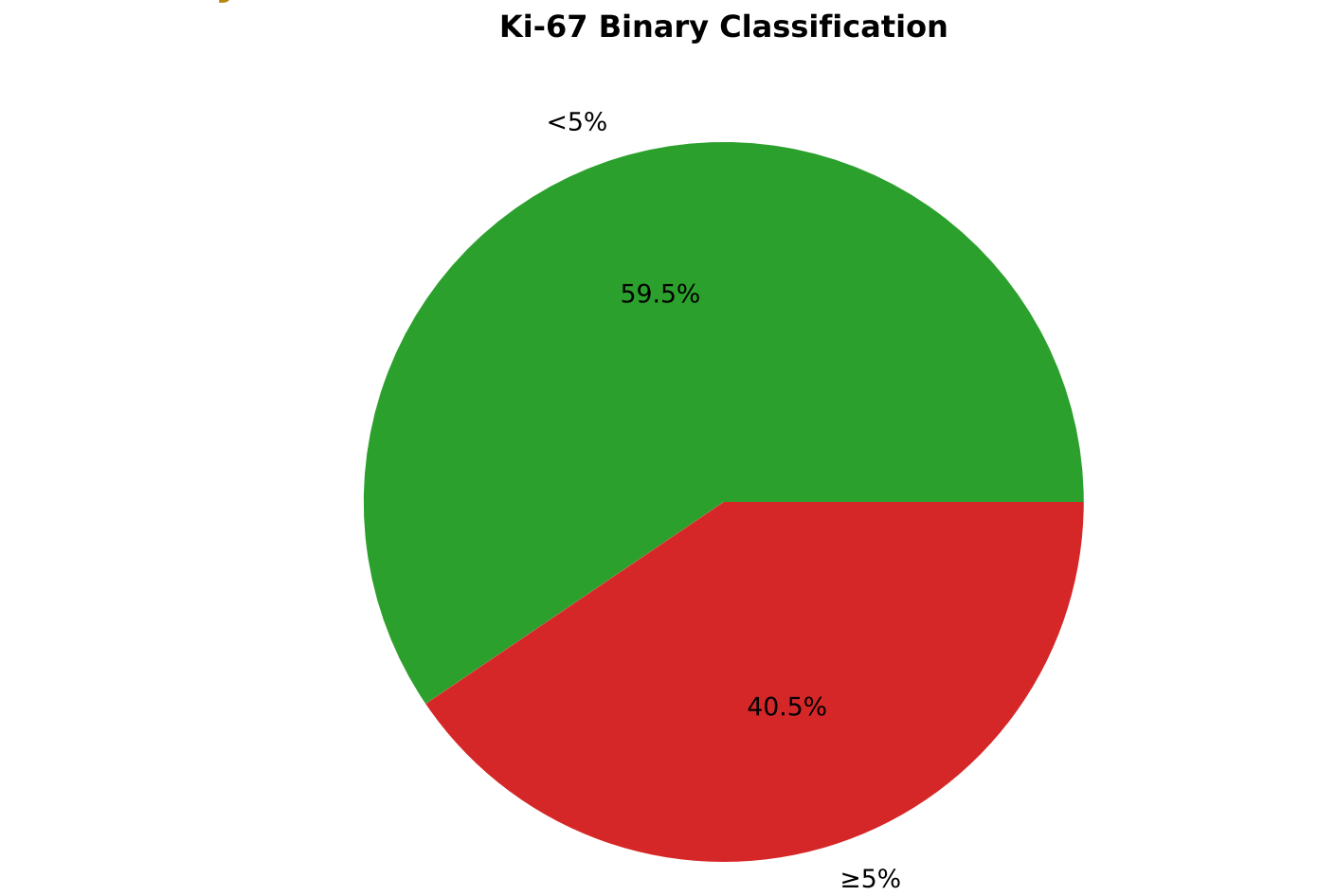
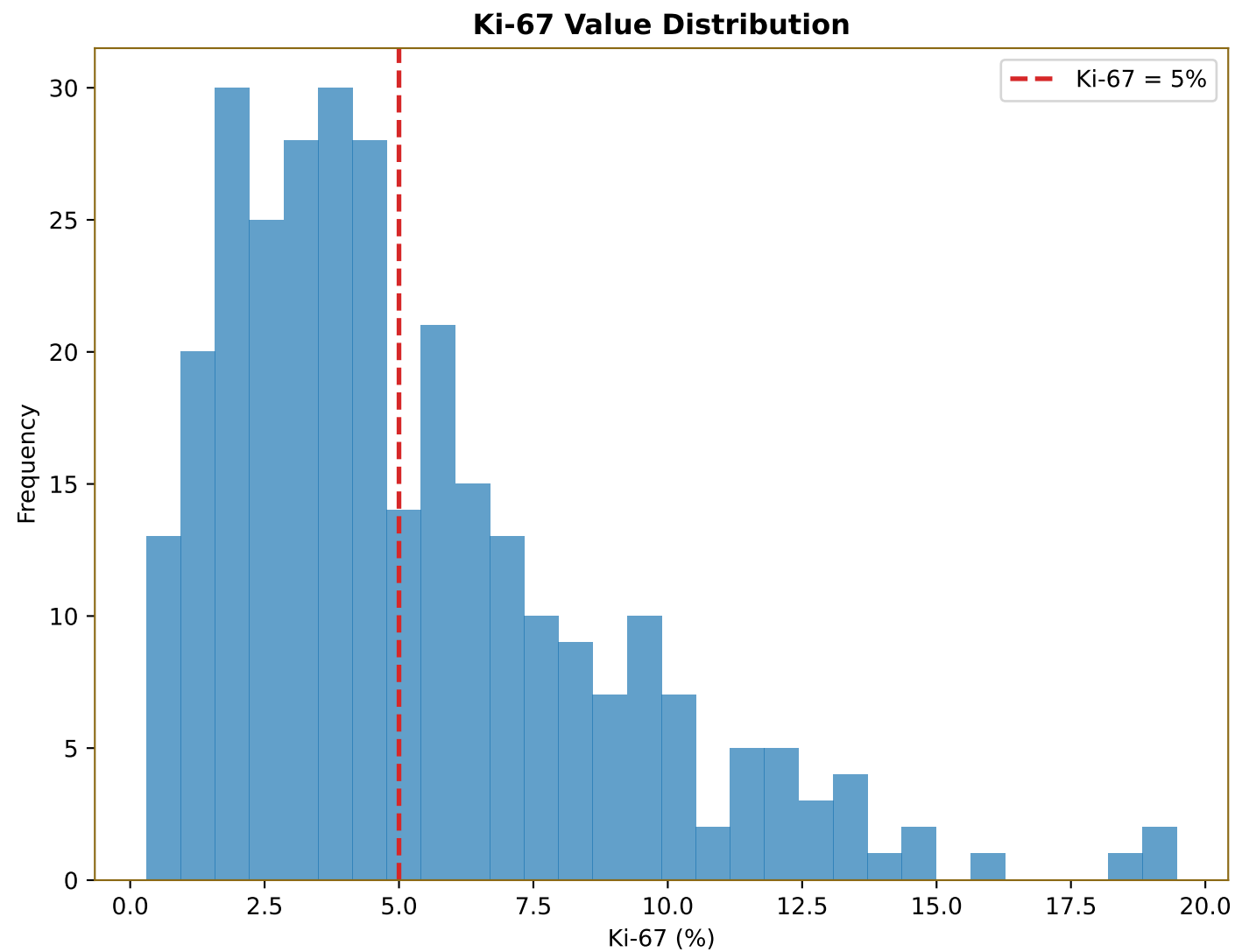
Tumor Laterality



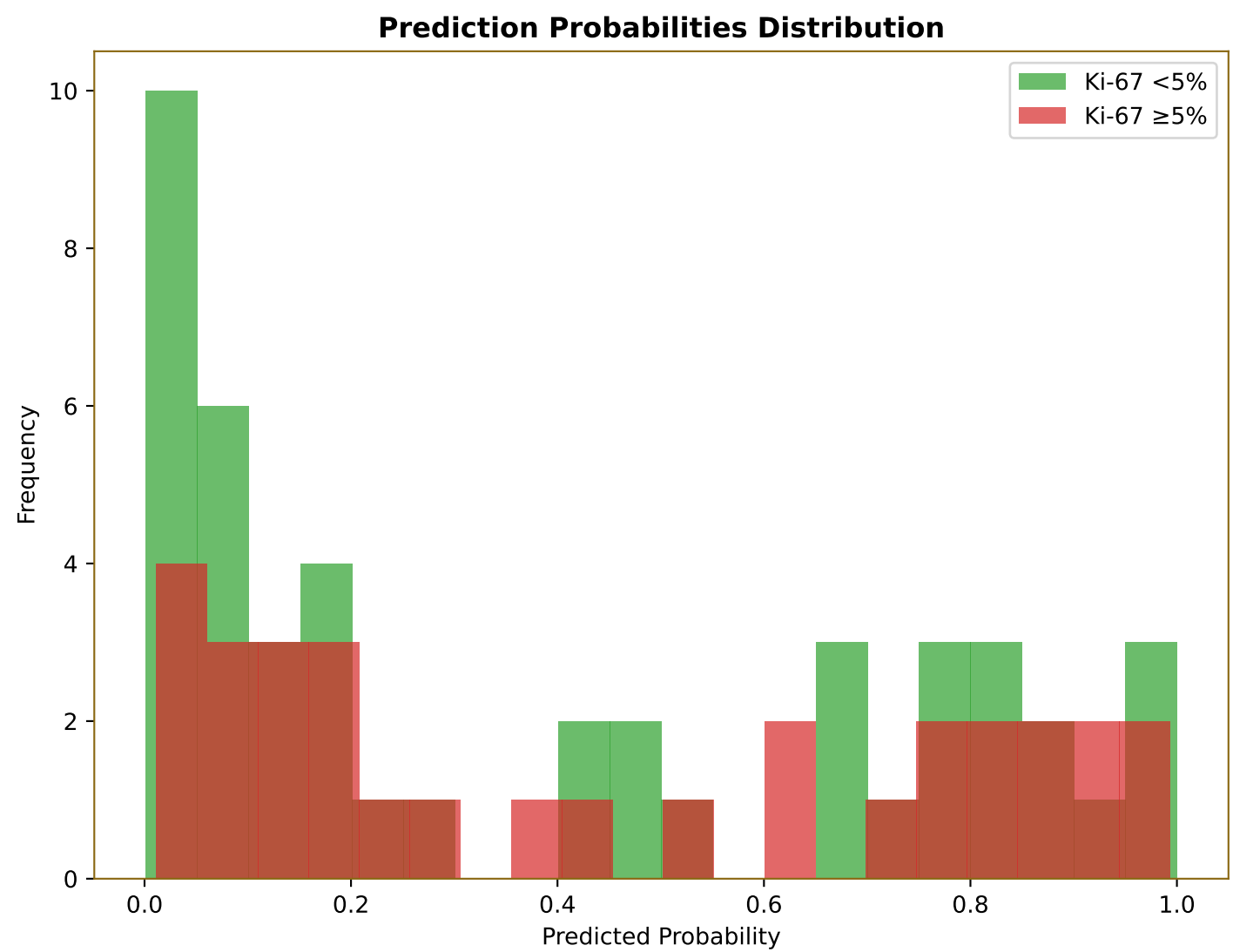
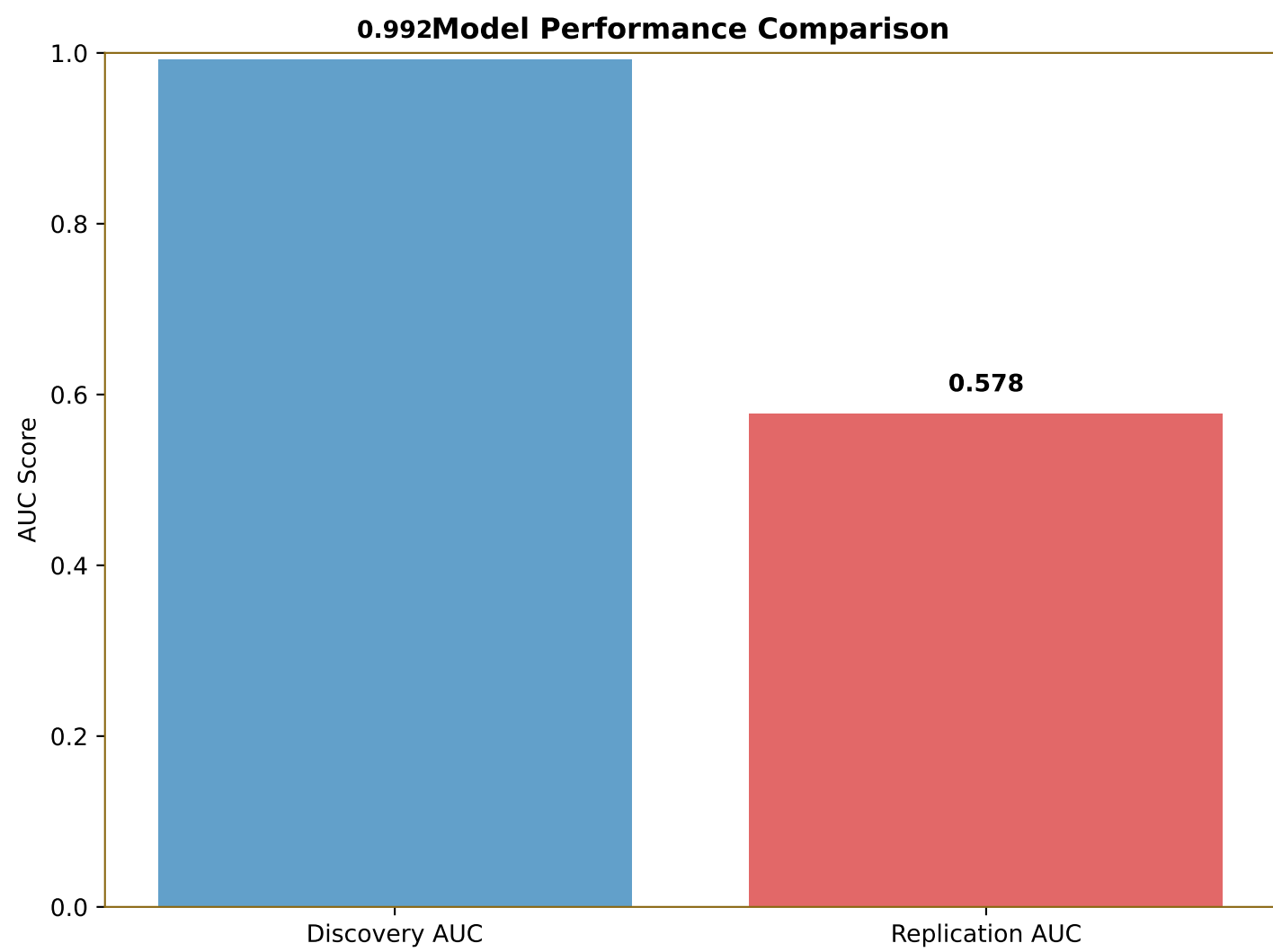
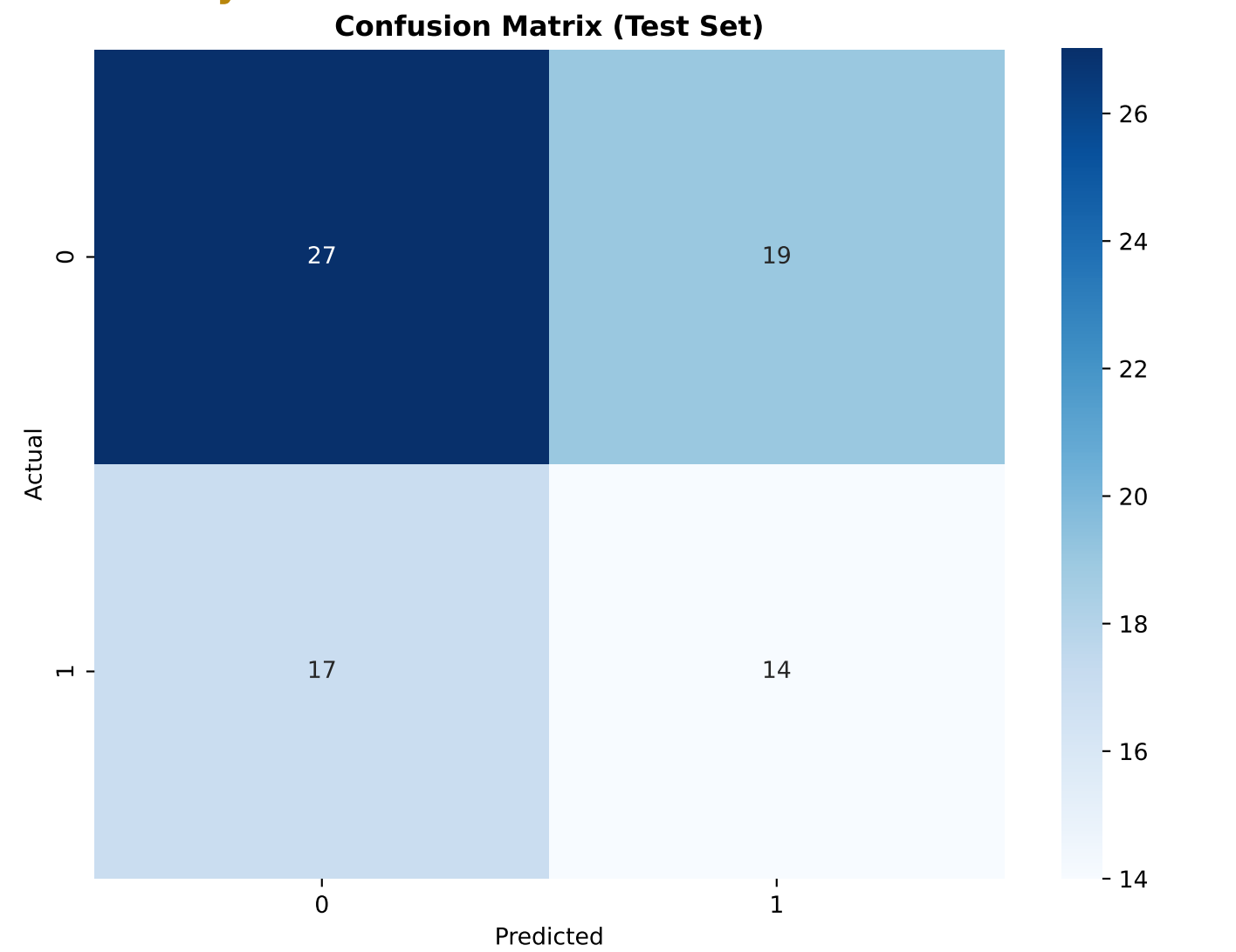
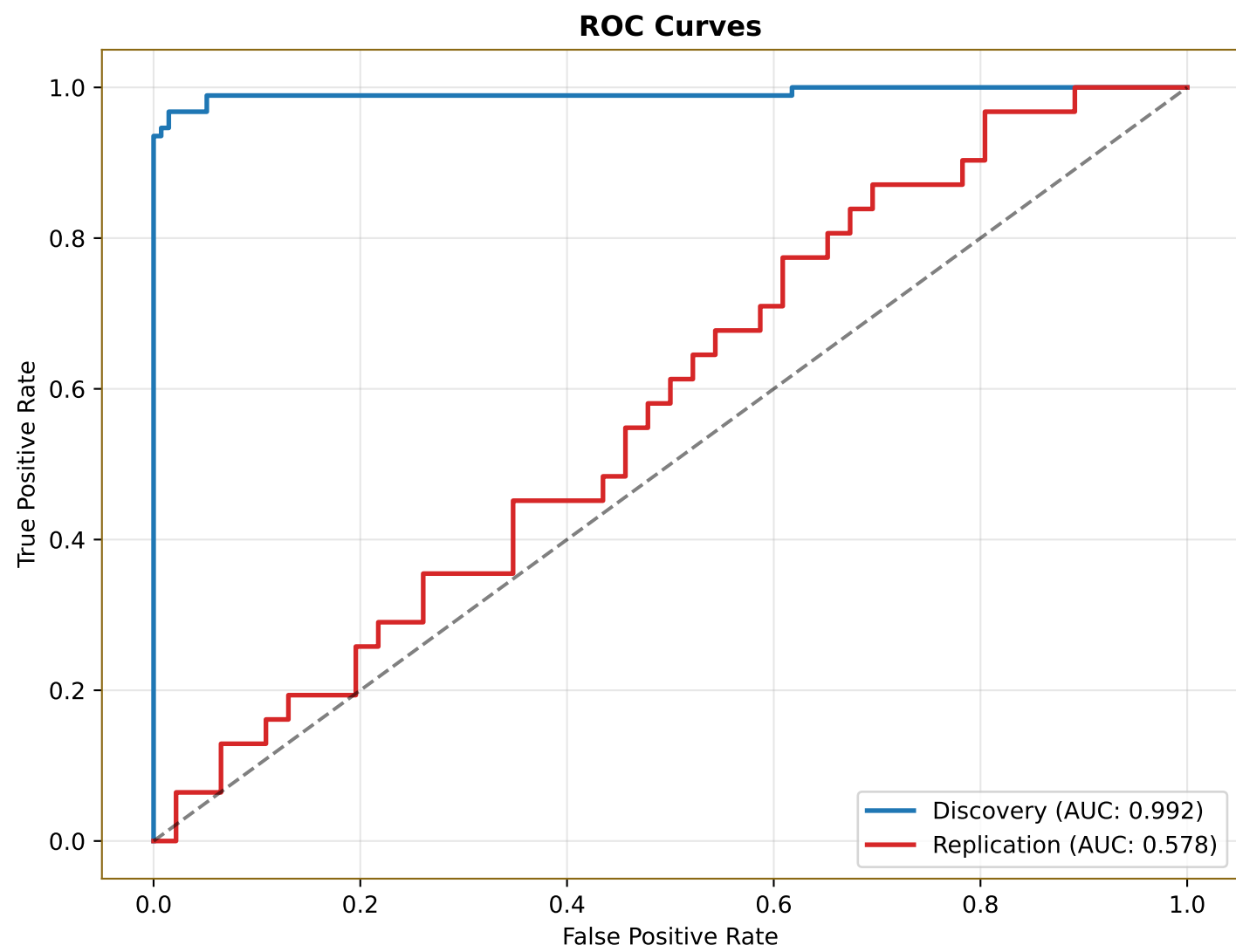
Skull Base vs Non-Skull Base



Ki-67 Distribution Analysis

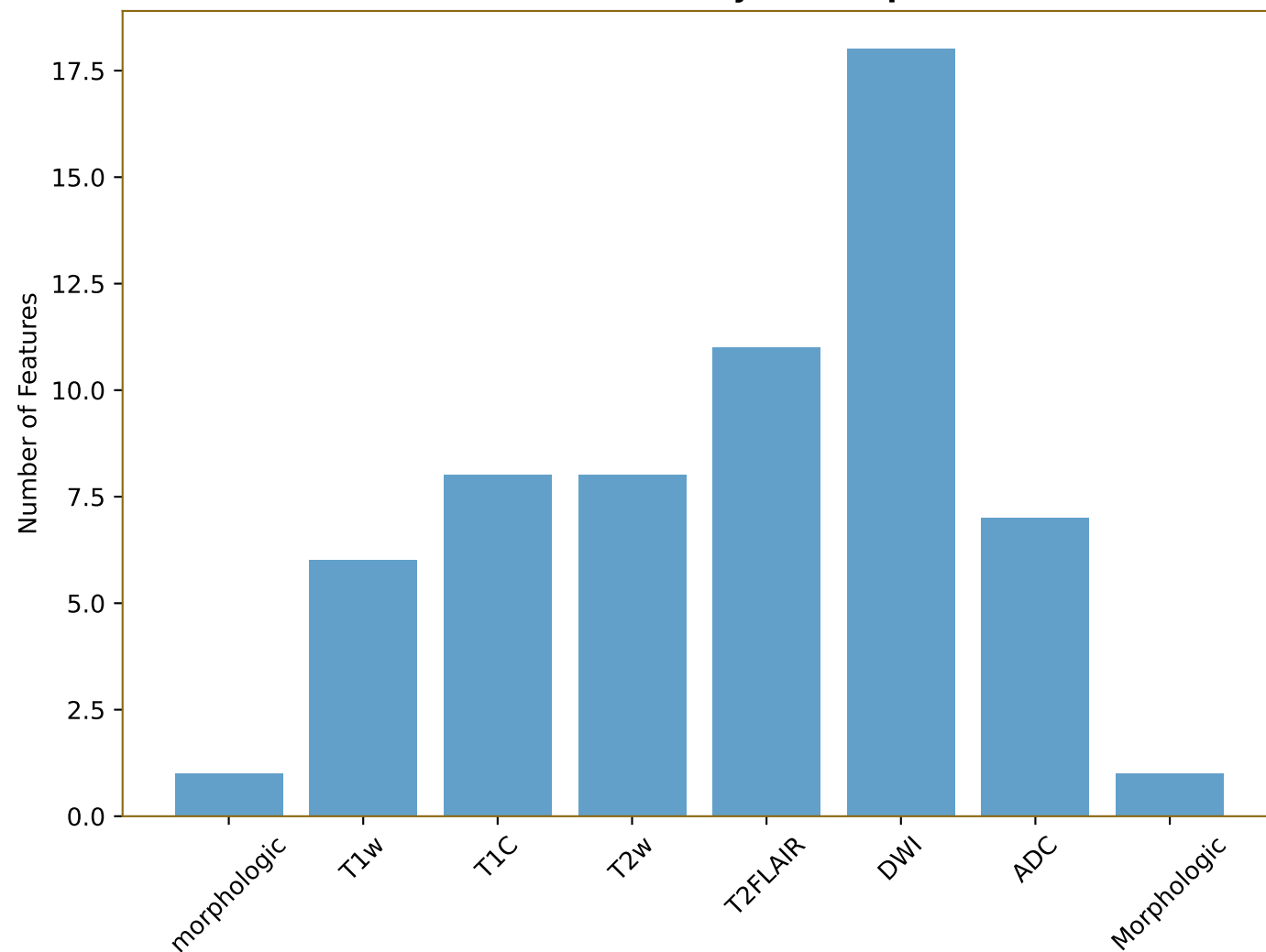


Model Performance Analysis

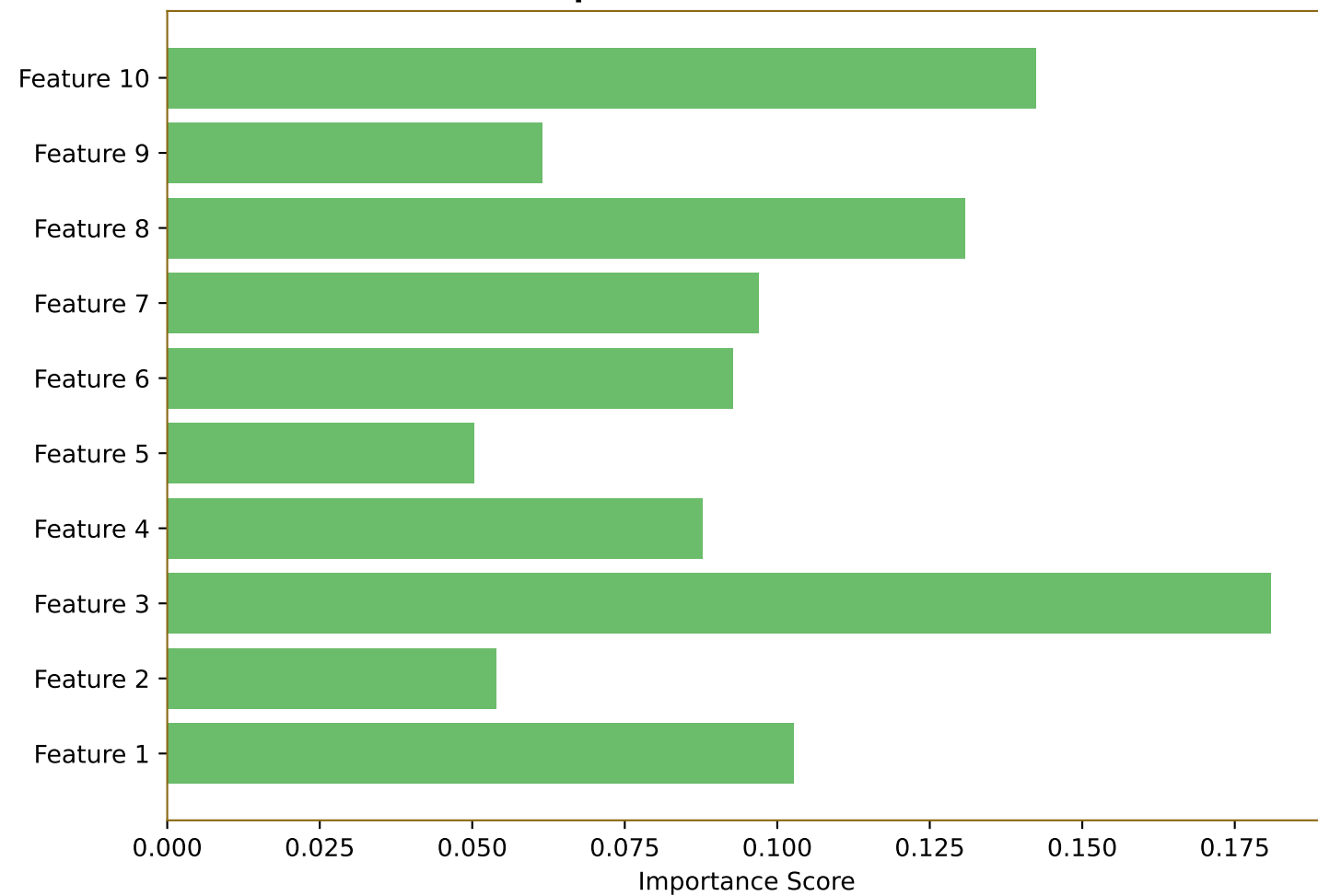


Feature Importance Analysis

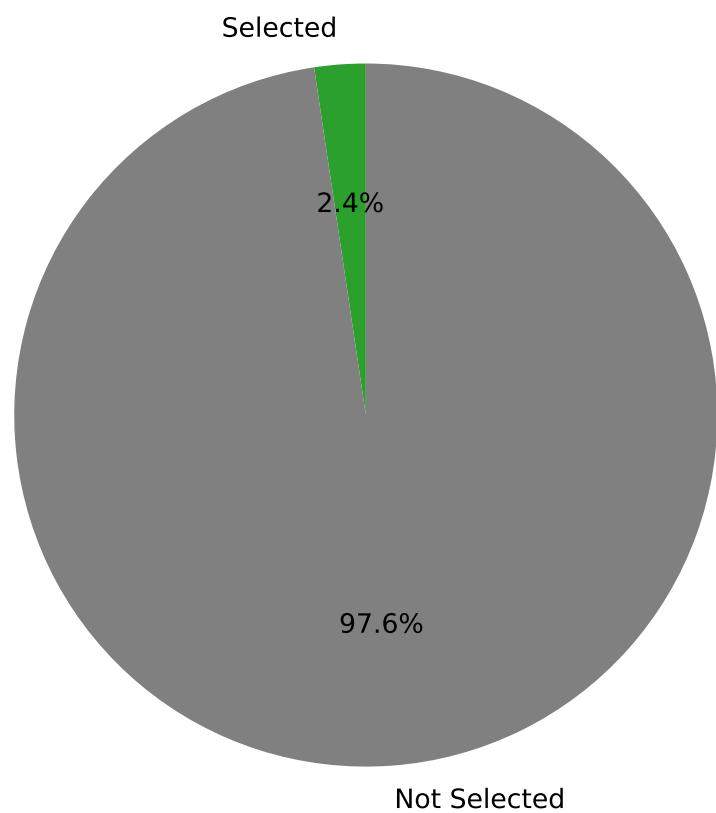
Selected Features by MRI Sequence



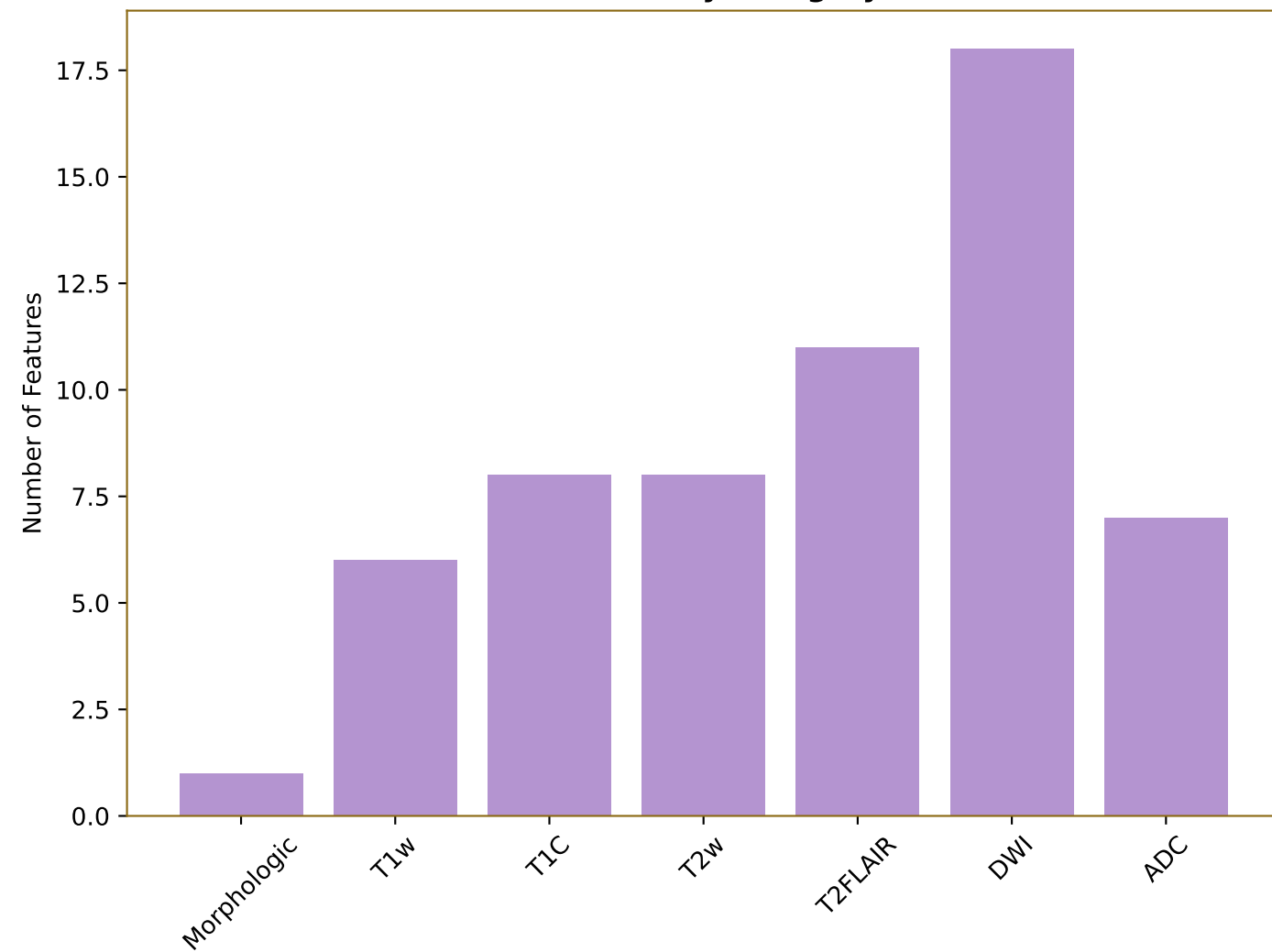
Top 10 Selected Features



Feature Selection Summary

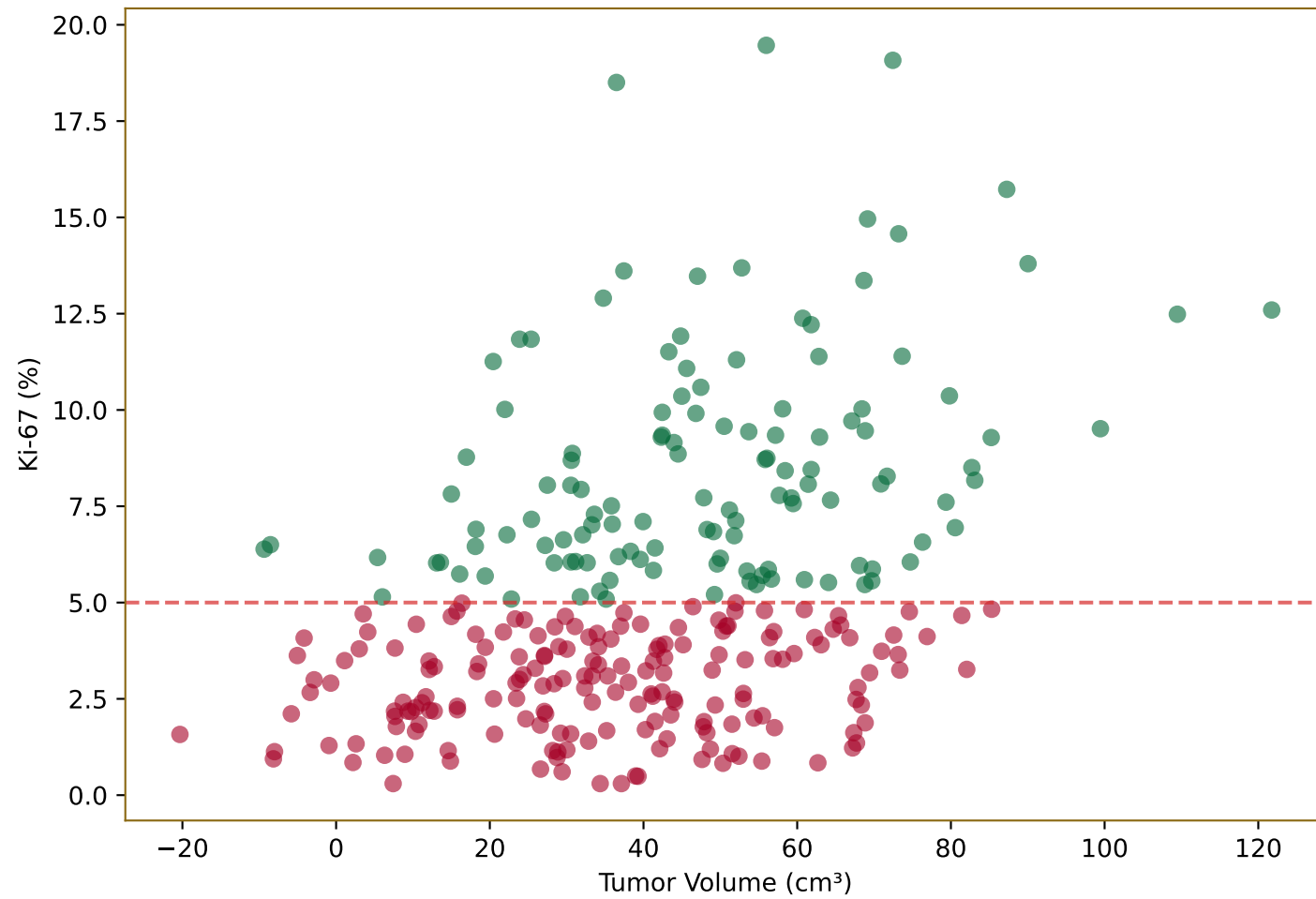


Features by Category

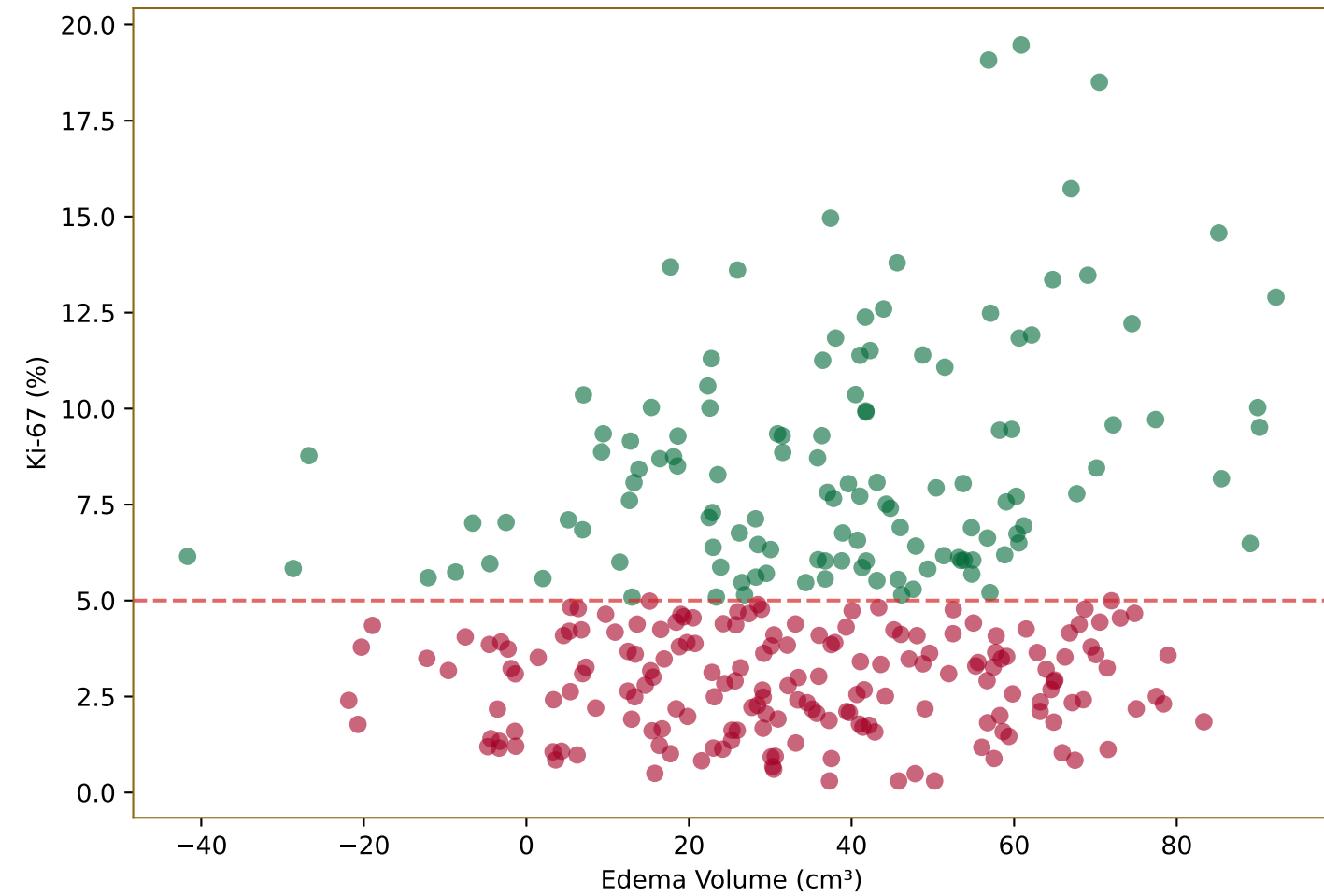


Tumor Characteristics Analysis

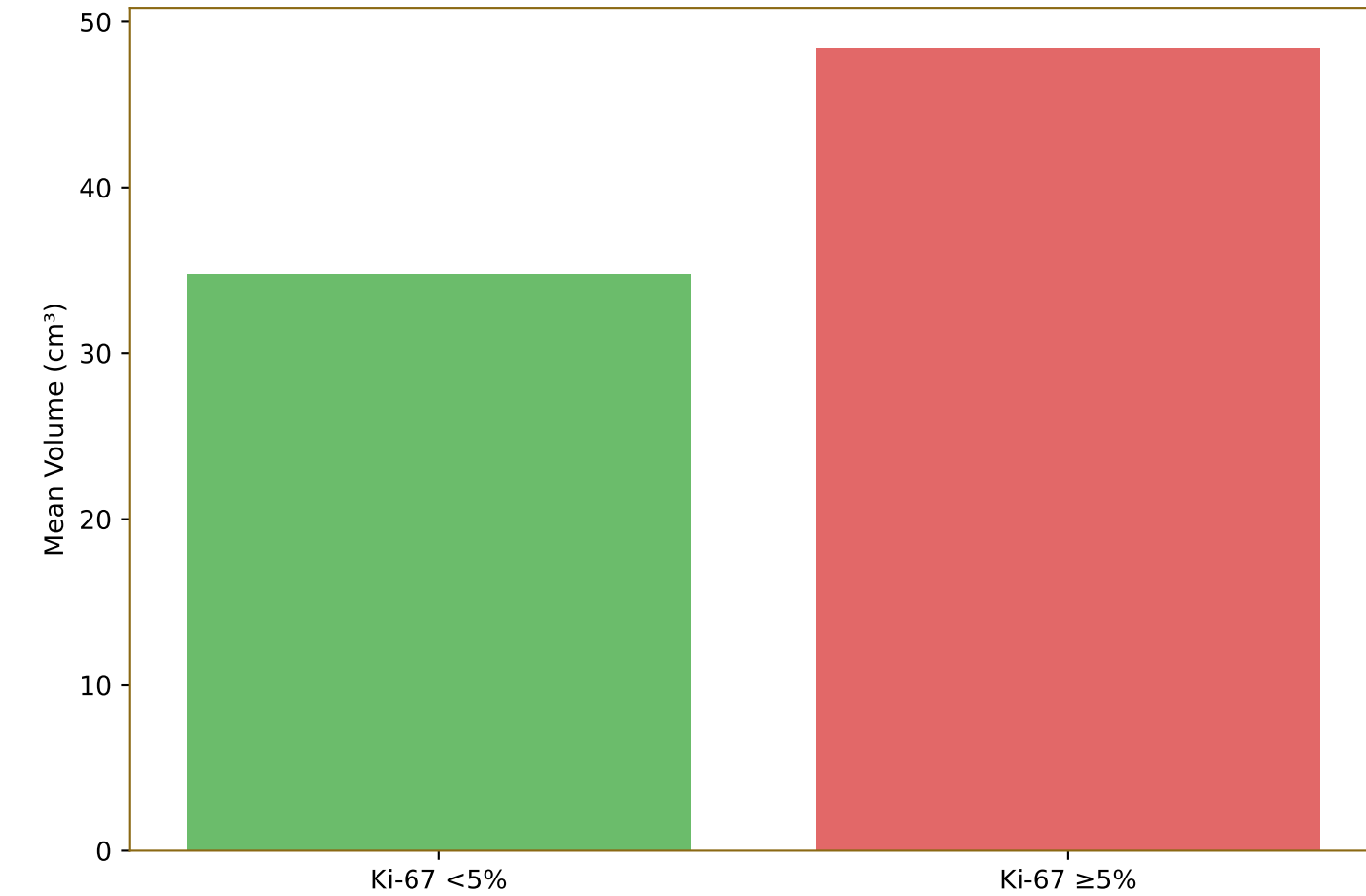
Tumor Volume vs Ki-67



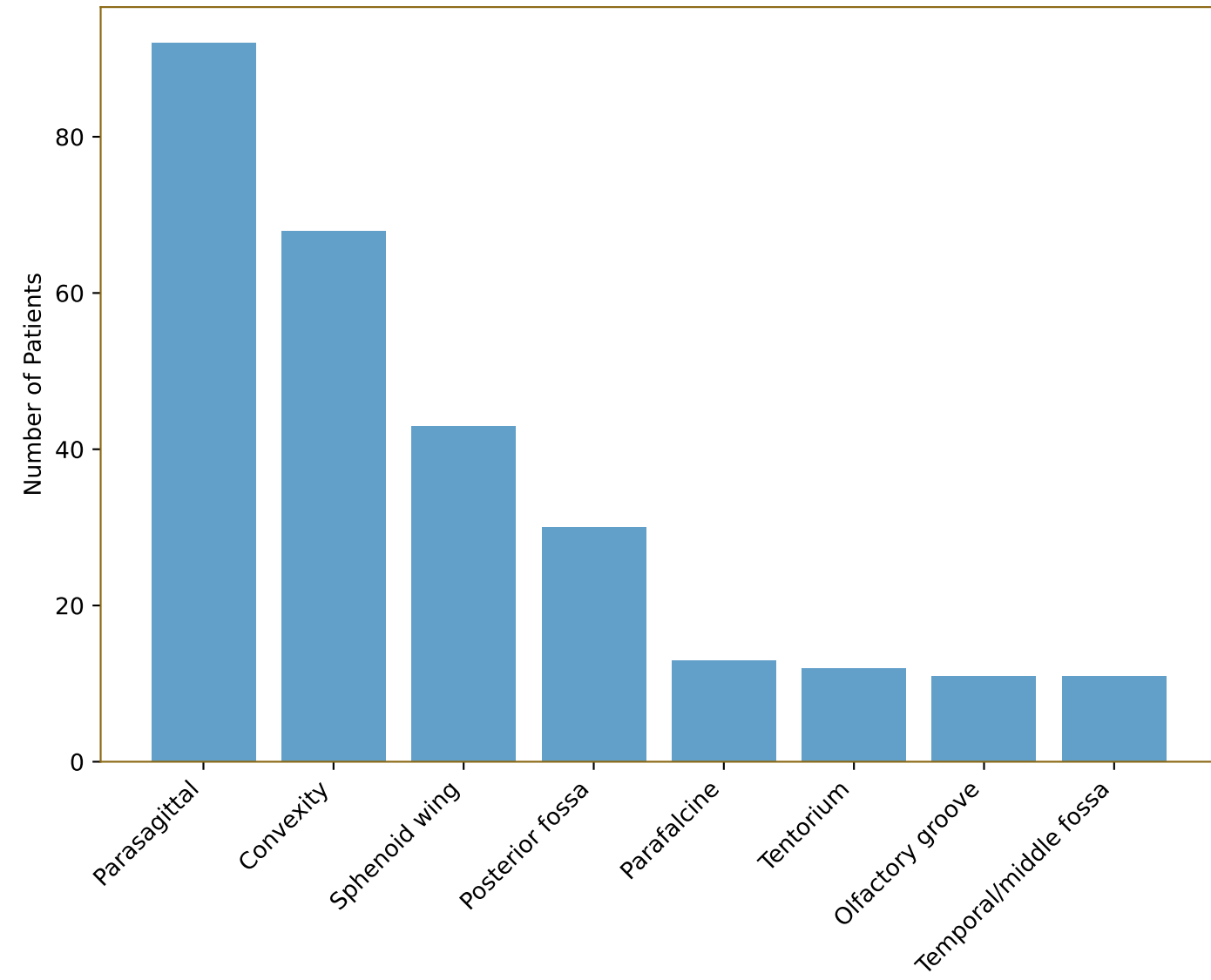
Edema Volume vs Ki-67



Mean Tumor Volume by Ki-67 Group



Tumor Locations



Summary & Clinical Implications

□ KEY RESULTS:

- Dataset: 306 patients with WHO grade I meningiomas
- Features: 2551 radiomic features extracted
- Selected Features: 60 features in final model
- Discovery Cohort AUC: 0.992
- Replication Cohort AUC: 0.578

□ CLINICAL IMPLICATIONS:

- Ki-67 Distribution: 182 <5%, 124 ≥5%
- Preoperative Ki-67 prediction guides surgical strategy
- Tumors with Ki-67 ≥5% require more aggressive resection
- Radiomics can identify aggressive grade I meningiomas
- Model applicable to both skull base and non-skull base tumors
- Automated analysis reduces inter-observer variability

□ METHODOLOGY HIGHLIGHTS:

- Multiparametric MRI (7 sequences) for comprehensive analysis
- LASSO feature selection for dimensionality reduction
- Support Vector Machine for robust classification
- Nested cross-validation for model optimization
- Independent replication cohort for validation

□ FUTURE DIRECTIONS:

- External validation on multi-institutional datasets
- Integration with clinical outcome prediction
- Real-time implementation in clinical workflow
- Extension to grade II and III meningiomas
- Longitudinal analysis for recurrence prediction